

Model-Assisted Analysis of Simultaneous Paraffin and Asphaltene Deposition in Laboratory Core Tests

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A model for analysis and interpretation of the simultaneous deposition of paraffin and asphaltene in laboratory core tests is presented. This model incorporates the mass balance equations for the oil, gas, paraffin, and asphaltene pseudocomponents; the paraffin and asphaltene solubility and deposition models; the energy and momentum balance equations; and the porosity and permeability reduction models. This model is numerically solved and validated with two sets of laboratory experimental data. It is demonstrated that the present model satisfactorily represents the simultaneous deposition of paraffin and asphaltene and the resultant damage caused on the porosity and permeability of porous media. The model is used for determination of the various process parameters with the help of the experimental data obtained by laboratory core tests. [DOI: 10.1115/1.1924466]

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Introduction

The organic deposition during oil production by primary depletion under isothermal conditions is preferentially of the asphaltene types. However, in some situations encountered in the field involving temperature variation, paraffin and asphaltene may precipitate and deposit together. For example, injection of fluids at temperatures colder than the reservoir oil temperature during hydraulic-fracturing or acid-stimulation processes may reduce the conditions of the reservoir oil to below the cloud point of paraffin. After the fracturing, the well begins to produce oil and the pressure in the near wellbore region declines. As a result of the simultaneous effect of temperature decrease and pressure depletion, paraffin and asphaltene may deposit in porous rock simultaneously in the near-wellbore region. For such applications, it is necessary to use a model for simultaneous deposition of paraffin and asphaltene in porous media.

Civan [1] first modeled and simulated the simultaneous deposition of paraffin and asphaltene. Wang et al. [2] presented an improved model to simulate the simultaneous deposition of paraffin and asphaltene in core tests. But the ideal solution theory was used to simulate the asphaltene and paraffin solubility, which may generate unrealistic results under nonideal conditions.

In this paper, a polymer solution theory is applied to simulate the dissolution-precipitation process of asphaltene in oil. A real solution model is used for paraffin dissolution and precipitation. The equations required for the paraffin and asphaltene deposition kinetics and the porosity and permeability reduction are also presented. The model is solved numerically and validated with two sets of experimental data.

Mass Balance Equations

The simultaneous paraffin and asphaltene deposition problem involves two fluid phases and four pseudocomponents. The two

phases are the vapor and oil phases, and the four pseudocomponents are gas, oil, paraffin, and asphaltene. The mass balance equations are written separately for each component. The mass balance equations for gas, oil, and asphaltene have been presented elsewhere for asphaltene deposition [3] and are not repeated here. Therefore, only the mass balance equation for paraffin is presented here.

The paraffin component can be dissolved in oil, suspended in oil as paraffin precipitate, or deposited in the pore space as solid deposits. Therefore, the mass balance equation of paraffin has the following form:

$$\frac{\partial}{\partial t}(\phi S_L C_P \rho_P + \phi S_L \rho_L w_{PL}) + \frac{\partial}{\partial x}(\rho_L u_L w_{SPL} + \rho_L u_L w_{PL}) = -\rho_P \frac{\partial E_P}{\partial t} \quad (1)$$

where C_P denotes the concentration (volume fraction) of paraffin precipitate in the oil phase. w_{PL} and w_{SPL} are the mass fractions of the dissolved paraffin and suspended paraffin precipitate in the oil. ρ_P is the density of paraffin. E_P is the volume fraction of paraffin deposits in bulk volume of porous rock.

Energy Balance Equation

Assuming that the oil, gas, paraffin, asphaltene, and rock matrix have the same temperature at a certain space, the total energy balance equation of the overall system at thermal equilibrium can be expressed as [2]

$$\begin{aligned} & \frac{\partial}{\partial t}[\phi S_V \rho_V H_V + \phi S_L \rho_L H_L + (\phi S_L C_P + E_P) \rho_P H_P + (\phi S_L C_A \\ & + E_A) \rho_A H_A + (1 - \phi - E_P - E_A) \rho_F H_F] + \nabla \cdot (\rho_V \mathbf{u}_V H_V \\ & + \rho_L \mathbf{u}_L H_L + \rho_L \mathbf{u}_L w_{SPL} H_P + \rho_L \mathbf{u}_L w_{SAL} H_A) \\ & = \nabla \cdot \{[\phi S_V K_V + \phi S_L K_L + (\phi S_L C_P + E_P) K_P \\ & + (\phi S_L C_A + E_A) K_A + (1 - \phi - E_P - E_A) K_F] \nabla T\} \quad (2) \end{aligned}$$

In this equation, H_V , H_L , H_P , H_A , and H_F denote the enthalpies of the vapor, oil, paraffin, asphaltene, and porous formation, respectively. K_V , K_L , K_P , K_A , and K_F are the thermal conductivities of

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the vapor, oil, paraffin, asphaltene, and porous formation, respectively. T is the equilibrium temperature of the overall system.

Solubility of Paraffin in Oil

The major factor affecting the solubility of paraffin in crude oil is temperature. A real solution model is applied to simulate the paraffin precipitation-dissolution processes. The solubility of paraffin in crude oil can be described adequately by [4]

$$X_{PL} = X_{PS} \exp \left[-\frac{\Delta H_P}{R} \left(\frac{1}{T} - \frac{1}{T_{PM}} \right) - \frac{V_P}{RT} (\delta_m - \delta_P)^2 - \ln \left(\frac{V_P}{V_m} \right) - 1 + \frac{V_P}{V_m} \right] \quad (3)$$

where X_{PL} represents the mole fraction of the paraffin dissolved in the oil as a function of temperature T . X_{PS} is the mole fraction of the paraffin precipitate in the total precipitates of paraffin and asphaltene, ΔH_P is the latent heat of fusion of the paraffin, R is the universal gas constant, T_{PM} is the melting point or fusion temperature for paraffin, and V_P and V_m are the molar volumes of paraffin and oil, respectively. δ_P and δ_m are the solubility parameters of paraffin and oil, respectively.

Solubility of Asphaltene

The precipitation and dissolution of asphaltene are represented by two equilibriums: the vapor-oil and oil-oil equilibriums. The vapor-oil equilibrium is first calculated with the Peng-Robinson equation. Then, the solubility of asphaltene in oil is calculated by a polymer solution model in oil-oil equilibrium. The details of the calculation and explanation are presented elsewhere by Wang and Civan [3,5].

Deposition Kinetics for Paraffin and Asphaltene The net deposition rate for asphaltene is expressed by [3]

$$\dot{E}_A = \alpha_A S_L C_A \phi - \beta_A E_A (v_L - v_{cr,L}) + \gamma_A u_L S_L C_A \quad (4)$$

The net deposition rate for paraffin is given by a similar expression as

$$\dot{E}_P = \alpha_P S_L C_P \phi - \beta_P E_P (v_L - v_{cr,L}) + \gamma_P u_L S_L C_P \quad (5)$$

In the above equations, it is assumed that the critical interstitial velocities of the oil carrying the paraffin and asphaltene precipitates are the same for both precipitates. The first term represents the surface deposition. The second term represents the entrainment of paraffin or asphaltene deposits by the flowing oil phase from the pore surface when the interstitial velocity exceeds the critical interstitial velocity required for sweeping the surface deposits. The last term represents the pore throat-plugging term. The above equations include the surface deposition coefficients α_A and α_P , the entrainment or pore surface-sweeping coefficients β_A and β_P , and the pore throat-plugging deposition coefficients γ_A and γ_P for asphaltene and paraffin, respectively.

The values of β_A and β_P are set as [6]

$$\beta_j = \beta_{ji}, \text{ when } v_L > v_{cr,L}, \quad j = P, A \quad (6)$$

$$\beta_j = 0, \text{ otherwise, } \quad j = P, A \quad (7)$$

where β_{ji} is the instantaneous entrainment rate coefficient, v_L is the interstitial velocity of oil phase equal to u_L/ϕ , and $v_{cr,L}$ is the critical interstitial velocity of the oil phase, which is considered to be constant. The values of γ_P and γ_A are determined according to the following pore throat-plugging criteria [6]:

$$\gamma_j = \gamma_{ji} [1 + \sigma_j (E_P + E_A)], \quad D_{pt} \leq D_{ptcr}, \quad j = P, A \quad (8)$$

$$\gamma_j = 0 \text{ otherwise, } \quad j = P, A \quad (9)$$

γ_{ji} is the instantaneous plugging deposition rate constant, σ_j represents σ_P or σ_A , both of which are constants, D_{pt} is the average

Table 1 Data used for simulation of the simultaneous paraffin and asphaltene deposition experiments of Sutton and Roberts [7]

Experiment	1	2
Temperature		
$T_m, ^\circ\text{C}$	54.4	54.4
$T_{oil}, ^\circ\text{C}$	21.1	21.1
Gas pseudo-component		
$M_g, \text{g/mole}$	16.0	16.0
$\rho_{gc}, \text{g/cm}^3$	0.00083	0.00083
Oil pseudo-component		
$M_o, \text{g/mole}$	104.11	122.51
$\rho_{oc}, \text{g/cm}^3$	0.72	0.75
Paraffin pseudo-component		
$M_p, \text{g/mole}$	522.4	478.7
$w_{ps} (\%)$	4.1	6.1
$\rho_{pc}, \text{g/cm}^3$	0.83	0.98
$T_{pm}, ^\circ\text{C}$	75.7	71.7
$\Delta H_p, \text{cal/g-mole}$	26,000	23,600
Asphaltene pseudo-component		
$M_A, \text{g/mole}$	2,500.0	2,500.0
$w_A (\%)$	0.7	0.1
$\rho_{Ac}, \text{g/cm}^3$	1.1	1.1
Core Properties		
L, cm	30.5	30.5
D, cm	2.5	2.5
$\phi, \text{fraction}$	0.25	0.25
k_p, Darcy	0.405	0.314
Experimental Conditions		
Bubble point pressure,	2050	2050
Flow rate, ft^3/day	0.38	0.30
Back pressure, psia	1900	1900

pore throat diameter, and D_{ptcr} is the critical pore throat diameter, assumed constant. Pore throat plugging occurs when D_{pt} is less than D_{ptcr} .

All deposition parameters mentioned in this section are estimated by the method of least-squares matching of experimental data using a trial-and-error approach.

Other Equations. The momentum balance equations, porosity and permeability reduction models, and other auxiliary equations are presented elsewhere in a previous paper [3].

Model Validation With Core Tests Involving Simultaneous Paraffin and Asphaltene Deposition. The numerical solution of the model equations described here was accomplished by means of the numerical method derived elsewhere [3]. The model has been validated using two sets of core test data of Sutton and Roberts [7] involving a simultaneous deposition of the paraffin and asphaltene obtained by cooling of the oil. The oil and core data, the time increments and grid-point spacing used in the numerical solution, and the best estimates of the parameter values are presented in Tables 1 and 2. The same values of the model parameters were assumed for both paraffin and asphaltene.

Sutton and Roberts [7] did not report the oil compositions and bubble-point pressures of crude oils. Therefore, the composition of the crude oil reported by Burke et al. [8] was used as an approximation and the bubble point pressures of crude oils were assumed as being 2050 psia. The value of the C_{7+} shift parameter was adjusted to match the asphaltene composition of the crude oils. The flow rates applied during the core tests were not reported by Sutton and Roberts [7]. However, the flow rate or velocity information is important for modeling of permeability impairment by organic deposition. The effect of velocity on the permeability decrease is based on two mechanisms, as considered in Eqs. (4) and (5). First, when the velocity of the stream is sufficiently high,

Table 2 Parameters used for simulation of the simultaneous paraffin and asphaltene deposition experiments of Sutton and Roberts [7]

Experiment	1	2
Simulation Parameters		
Grid spacing, Δx , cm	2.54	2.54
Time increment, Δt , sec	40.0	40.0
Number of blocks	12	12
Deposition Parameters		
D_{ptcr} , cm	N/A	N/A
$v_{c,l}$, cm/sec	N/A	N/A
$\alpha_p = \alpha_A$, 1/sec	0.0241	0.0182
$\beta_{pt} = \beta_A$, 1/cm	0.0	0.0
$\gamma_{pt} = \gamma_A$, 1/cm	0.0	0.0
$\sigma_p = \sigma_A$, constant	0.0	0.0
Deposition mechanism	Surface deposition	Surface deposition

the stream may entrain some asphaltene and paraffin deposits from the pore surface by sweeping. Second, the pore throat-plugging effect is enhanced at high stream velocity because the probability of the migrating particles to bridge across the pore throats increases. Based on the parameters obtained in matching the data of Sutton and Roberts [7] in this study, only surface deposition occurs in both cases, and both of entrainment of deposits and pore throat plugging do not occur. Ring et al. [9] used 0.38 ft³/day and 0.30 ft³/day for their simulation of the Experiment 1 and 2 data of Sutton and Roberts. Thus, the same flow rates were also used in the present study. The backpressure is taken as 1900 psia for both tests. However, Ring et al. [9] only considered the paraffin deposition.

Experiment 1. Sutton and Roberts [7] saturated a Berea sandstone core with a Shannon Sand crude oil first and then heated the oil-saturated core to 54.4°C without flowing oil. Second, the core outlet side was cooled down to 21.1°C for 2 h without flowing oil. Third, the Shannon Sand crude oil was injected into the core at 54.4°C temperature. During the flow experiments, the temperature of the core outlet side was maintained at 21.1°C. This temperature is 16.7°C below the cloud point temperature of the oil. The Shannon Sand crude oil contains 4.1% (weight) paraffin and 0.7% (weight) asphaltene. The cloud point of the oil is 37.8°C. A set of 12 grid blocks was used in the simulation similar to Ring et al. [9]. Other data used in the simulation are provided in Tables 1 and 2. The simulation results are satisfactory as depicted in Fig. 1.

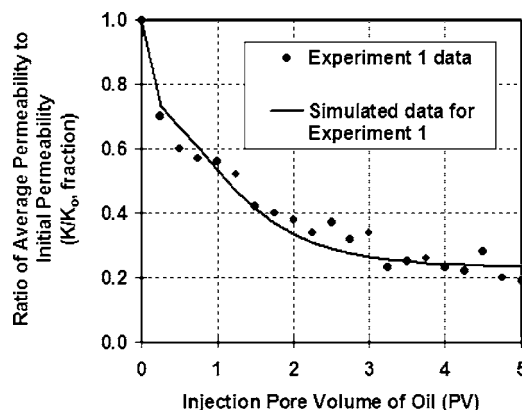


Fig. 1 Comparison of the simulation results with the Experiment 1 data of Sutton and Roberts [7]

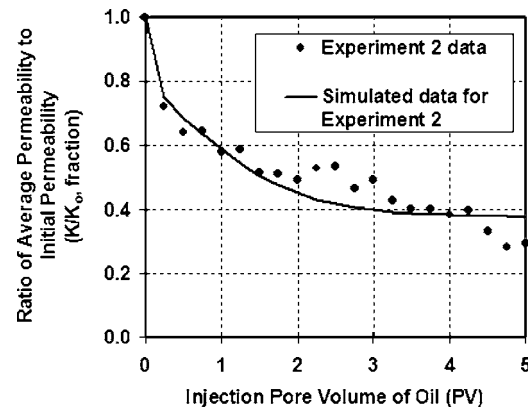


Fig. 2 Comparison of the simulation results with the Experiment 2 data of Sutton and Roberts [7]

The best estimate values of the model parameters are given in Table 2. In this case, the primary deposition mechanism is surface deposition.

Experiment 2. Sutton and Roberts [7] carried out a second test by flowing Muddy formation crude oil through a Berea sandstone core. Also in this test, the outlet temperature of the core was kept at 21.1°C. This temperature is 13.9°C below the cloud point temperature (35°C) of the oil. This oil sample contained 6.1% (weight) paraffin and 0.1% (weight) asphaltene. The other data used in the simulation are presented in Table 1. The simulation results are satisfactory as depicted in Fig. 2. The best estimate values of the model parameters are given in Table 2. Again, the primary deposition mechanism is surface deposition.

Final Remarks and Conclusions

This study has shown that the simultaneous deposition of paraffin and asphaltene in porous media may cause serious formation damage. The model and approach presented here can be applied for model-assisted analysis of the simultaneous deposition of paraffin and asphaltene in laboratory core tests and simulation of the consequences of organic deposition in porous media.

Nomenclature

- C_p = paraffin concentration in oil phase, volume fraction
- D_{pt} = mean diameter of pore throats, cm
- D_{ptcr} = critical diameter of pore throats, cm
- E_A = volume fraction of deposited asphaltene, dimensionless
- E_p = volume fraction of deposited paraffin, dimensionless
- H_V = enthalpy of vapor phase, cal/cm³
- H_L = enthalpy of oil phase, cal/cm³
- H_p = enthalpy of paraffin, cal/cm³
- H_A = enthalpy of asphaltene, cal/cm³
- H_F = enthalpy of formation, cal/cm³
- ΔH_p = latent heat of fusion of paraffin, cal/g mol
- K_V = thermal conductivity of vapor, cal/s cm K
- K_L = thermal conductivity of oil, cal/s cm K
- K_p = thermal conductivity of paraffin, cal/s cm K
- K_A = thermal conductivity of asphaltene, cal/s cm K
- K_F = thermal conductivity of formation, cal/cm K
- M_g = molecular weight of gas, g/g mol
- M_o = molecular weight of oil, g/g mol
- M_p = molecular weight of paraffin, g/g mol
- M_A = molecular weight of asphaltene, g/g mole
- R = universal gas constant, 1.987 cal/(mol K)

Q = flow rate, cm^3/s
 S_V = saturation of vapor phase, dimensionless
 S_L = oil saturation, fraction
 t = time independent variable, sec
 T = temperature, $^{\circ}\text{F}$ or $^{\circ}\text{R}$
 T_{PM} = melting point temperature of paraffin, $^{\circ}\text{C}$
 u = superficial velocity, cm/s
 u_V = velocity of vapor, cm/s
 u_L = velocity of oil phase, cm/s
 v_{cr} = critical interstitial velocity for entrainment, cm/s
 v_L = interstitial velocity of oil phase, cm/s
 V_P = molar volumes of paraffin, m^3/kmol
 V_m = molar volumes of oil, m^3/kmol
 w_{PL} = mass fraction of dissolved paraffin in the oil phase, dimensionless
 w_{PS} = initial weight fraction of paraffin in the oil phase, dimensionless
 w_{SPL} = the mass ratio of the paraffin precipitates suspended in oil to the oil phase, dimensionless
 x, y, z = Cartesian space variables, cm
 X_{PL} = mole fraction of paraffin dissolved in oil, dimensionless
 X_{PS} = mole fraction of the precipitated paraffin of the total volume of the precipitated paraffin and asphaltene, dimensionless

Greek

ρ_V = density of vapor, g/cm^3
 ρ_L = density of oil phase, g/cm^3
 ρ_P = density of paraffin, g/cm^3
 ρ_A = density of asphaltene, g/cm^3
 ρ_F = density of formation, g/cm^3
 α_P = surface deposition rate coefficient for paraffin, $1/\text{s}$
 β_P = entrainment rate coefficient, $1/\text{cm}$
 β_{Pi} = instantaneous entrainment rate coefficient, $1/\text{cm}$
 γ_P = plugging deposition rate coefficient for paraffin, $1/\text{s}$
 γ_{Pi} = instantaneous plugging deposition rate coefficient for paraffin, $1/\text{s}$
 σ_P = pore-throat plugging rate coefficient for paraffin, constant
 γ_A = plugging deposition rate coefficient for asphaltene, $1/\text{s}$

γ_{Ai} = instantaneous plugging deposition rate coefficient for asphaltene, $1/\text{s}$
 σ_A = pore-throat plugging rate coefficient for asphaltene, constant
 δ_P = solubility parameters of paraffin, $(\text{MPa})^{1/2}$
 δ_m = solubility parameters of oil, $(\text{MPa})^{1/2}$

Subscripts

A = asphaltene
 cr = critical
 e = entrainment
 F = formation
 G = gas
 i = instantaneous
 L = oil
 M = melting point
 O = oil
 p = plugging deposition
 P = paraffin
 pt = pore throat
 V = vapor

References

- [1] Civan, F., 1995, "Modeling and Simulation of Formation Damage by Organic Deposition," *Proc. of First Int. Symp. in Colloid Chemistry in Oil Production: Asphaltene and Wax Deposition*, ISOP'95, Rio de Janeiro, Nov. 26–29, pp. 102–107.
- [2] Wang, S., Civan, F., and Strycker, A. R., 1999, "Simulation of Paraffin and Asphaltene Deposition in Porous Media," *SPE 1999 Int. Symp. on Oilfield Chemistry*, Houston, Feb. 16–19, Paper No. SPE 50746, pp. 57–66.
- [3] Wang, S., and Civan, F., 2005, "Modeling Formation Damage by Asphaltene Deposition During Primary Oil Recovery," *ASME J. Energy Resour. Technol.*, in print.
- [4] Chung, T., 1992, "Thermodynamic Modeling for Organic Solid Precipitation," *67th Annual Technical Conference and Exhibition of the Society of Petroleum Engineers*, Washington, DC, Oct. 4–7, Paper No. SPE 24851.
- [5] Wang, S., and Civan, F., 2005, "Preventing Asphaltene Deposition in Oil Reservoirs by Early Water Injection," presented at the *2005 SPE Production and Operations Symposium*, Oklahoma City, April 17–19, No. SPE 94268.
- [6] Wang, S., and Civan, F., 2001, "Productivity Decline of Vertical and Horizontal Wells by Asphaltene Deposition in Petroleum Reservoirs," *SPE 2001 Int. Symp. on Oilfield Chemistry*, Houston, Feb. 13–16, Paper No. SPE 64991.
- [7] Sutton, G. D., and Roberts, L. D., 1974, "Paraffin Precipitation During Fracture Stimulation," *JPT*, **26**(9), pp. 997–1004.
- [8] Burke, N. E., Hobbs, R. E., and Kashon, S. F., 1990, "Measurement and Modeling of Asphaltene Precipitation," *JPT*, **42**(11), pp. 1440–1446.
- [9] Ring, J. N., Wattenbarger, R. A., Keating, J. F., and Peddibhotla, S., 1994, "Simulation of Paraffin Deposition in Reservoirs," *SPEPF*, **9**(1), pp. 36–42.



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